

■ Complete Hardware Components List

Scuba Diver Health Monitoring System — All Components

Hardware Components Table

#	Component	Model / Spec	Qty	Module / Purpose
1	ESP32 Microcontroller	Dual-core Xtensa LX6, 240MHz, Wi-Fi+BT	1	Transmitter & Receiver
2	Pulse Oximeter Sensor	MAX30102 (SpO2 + Heart Rate)	1	Oxygen & pulse monitoring
3	Heartbeat Sensor	KY-039 or MAX30102 PPG	1	Heart rate detection
4	Temperature Sensor	DS18B20 Waterproof, -55 to 125°C	1	Body temperature
5	Li-Fi Transmitter	LED + LM741 pre-amp + power amp	1	Underwater data TX
6	Li-Fi Receiver	Photodiode/LDR + LM358 amplifier	1	Underwater data RX
7	LCD Display	16x2 with I2C backpack module	2	Diver + surface display
8	Relay Module	5V 1-channel, SRD-05VDC	1	Controls air motor
9	Air Motor (Pneumatic)	Compressed air motor for balloon	1	Emergency ascent
10	Buzzer	5V active buzzer	2	Alert at surface + diver
11	Panic Switch	Waterproof push button (IP67)	1	Manual SOS signal
12	Power Supply	7805 + LM317 voltage regulators	1	+5V and +3.8V supply
13	Bridge Rectifier	WM04, 1A	1	AC to DC conversion
14	Capacitors	1000µF, 100nF	As needed	Filter capacitors
15	Resistors	220Ω, 4.7kΩ, various	As needed	Current limiting
16	Transformer	230V to 12-0-12V, stepped down	1	AC power input
17	Jumper Wires	M-M, M-F, 20cm	30+	Connections
18	Breadboard	830-tie point	2	Prototyping

Key Component Specifications

ESP32 Microcontroller

- Processor: Dual-core Xtensa LX6, 160–240 MHz
- Wi-Fi: 802.11 b/g/n | Bluetooth: Classic + BLE
- GPIO: 30+ pins (ADC, DAC, PWM, SPI, I2C, UART, CAN, Touch)
- Flash: 4–16 MB | RAM: 520 KB SRAM
- Power: 3.3V, Deep sleep mode supported
- IDE: Arduino IDE, MicroPython, ESP-IDF compatible

MAX30102 Pulse Oximeter Sensor

- Measures: SpO2 (Blood Oxygen) and Heart Rate (BPM)
- Interface: I2C (SDA, SCL)
- Operating voltage: 1.8V–3.3V
- Uses PPG (Photoplethysmography) with red and IR LEDs
- Accuracy: SpO2 $\pm 2\%$, Heart Rate ± 3 BPM
- Built-in 16-bit ADC for high resolution reading

Li-Fi Transmitter Module

- Two-stage circuit: Pre-amplifier (LM741) + Power Amplifier
- Converts electrical data signals into modulated LED light
- LED ON = digital 1 | LED OFF = digital 0
- Can be switched at very high speeds for data modulation
- Avoids RF interference; works underwater using visible light
- Suitable for short-range (2–3m) underwater communication

Li-Fi Receiver Module

- Two-stage circuit: Pre-amplifier (LM358) + signal output
- Uses LDR (Light Dependent Resistor) or Photodiode to receive light
- Converts optical signals back into electrical signals
- Voltage divider using R1 & R8 biases the LDR
- Solar cell can also be used as the photodetector element
- Low resistance in light → signal detected; high in dark → no signal

DS18B20 Waterproof Temperature Sensor

- Temperature range: -55°C to $+125^{\circ}\text{C}$
- Accuracy: $\pm 0.5^{\circ}\text{C}$ (-10°C to $+85^{\circ}\text{C}$ range)
- Interface: 1-Wire digital protocol (single data line)
- Waterproof stainless steel probe for underwater use
- Powered by 3.0V–5.5V
- Multiple sensors can share one data line